

ABDUL WAHAB

MECHANICAL DESIGN ENGINEER AND PRODUCT DESIGNER

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Portfolio: <https://wahababdull.github.io/CV/>

Research Interests: Autonomous and intelligent vehicle systems, robotic navigation and SLAM, AI applications in robotics, scene understanding for autonomous driving, end-to-end autonomous systems, innovative product and mechanical design, and design for manufacturability in robotic platforms.

Mechanical Design Engineer and Product Designer with a rigorous, methodology-driven approach to SolidWorks-based CAD modeling. Adept at transforming complex concepts into production-ready designs through creative problem-solving techniques, strict adherence to industry standards, and a deep understanding of manufacturing processes. My Design Philosophy is very usefull. I approach mechanical design with a researcher's mindset: prioritizing empirical validation, functional performance, and material efficiency. Every model is engineered not just for aesthetic appeal, but for absolute manufacturability and operational resilience.

SKILLS AND EXPERTISE

- CAD and Manufacturing Design
- Design Simulation and Manufacturability Analysis
- Manufacturing Process Design and Design for Manufacturability (DFM)
- Engineering Problem-Solving and Root Cause Analysis
- Process Data Analysis

PROFESSIONAL EXPERIENCE

Mechanical Design Engineer | R&D Department | Dysol (Pvt) Ltd, Lahore

June 2026 – Present

Working on three confidential product-development projects across the fitness and medical fields, involving mechanical mechanisms, fluid systems, and mechanical-electronic integration.

- Responsible for 3D CAD modeling, mechanism design, assembly development, DFM/DFA, design optimization, and iterative engineering development.
- Worked on complex systems involving gear/rack mechanisms, valves, pumps, fluid-control components, actuators, connectors, sensors, and other mechanical interfaces.
- Addressed challenges related to multi-function mechanisms, limited space/travel, pressure & flow control, tolerances, aesthetics, and subsystem integration through iterative design and engineering validation.
- **Skills:** SolidWorks , ANSYS , Altair EDEM , 2D Technical Drawing, DFM/DFA, Iterative Design and Engineering Validation.

Mechanical Design Engineer | Freelance (Upwork)

August 2025 – Present

- Delivered professional **mechanical and product design services** to international clients via **Upwork**.
- Created 3D CAD models, assemblies, 2D technical drawings, and photorealistic renders for prototyping and manufacturing.
- Designed components for consumer products and industrial applications, optimized for material selection and manufacturing processes.
- Communicated directly with clients to gather requirements, manage revisions, and meet project deadlines
- Maintained high client satisfaction through **quality delivery and clear technical communication**.
- **Skills:** SolidWorks, 3D CAD Modeling, 3D CAD Rendering, 2D Technical Drawing, Clear Communication, Detail Oriented Design, Prototyping Support

EDUCATION

Bachelor of Science | University of Engineering and Technology, Peshawar | Khyber Pakhtunkhwa, Pakistan

September 2019 – October 2023

- **Field:** Mechanical Engineering
- **Supervisor:** Professor Usman
- **Final Project:** Design and Fabrication of Mechanism for Automated Installation of 18 Liter water bottle of Water Dispenser

Faculty of Science | WENSAM college, Dera Ismail Khan | Khyber Pakhtunkhwa, Pakistan

September 2017 – June 2018

- **Field:** Pre-Engineering

PROJECTS

1. Fitness Product Development — Mechanical Dispensing System

Industry: Fitness & Consumer Products

Key Contributions:

- Designed mechanical assemblies involving gears, rack mechanisms, springs, sliding mechanisms, and functional interfaces.
- Developed solutions for multi-function operation within limited mechanical movement constraints.
- Performed multiple design iterations to improve mechanism reliability and manufacturability.
- Applied DFM and DFA principles for production-oriented design.
- Conducted motion analysis and mechanical validation.
- **Challenges Managed:** Optimizing complex mechanical interactions while maintaining compact design requirements. Achieved functional mechanism performance through iterative CAD refinement and simulation.
- **Tools Used:** SolidWorks (3D Modeling, Assembly Design, Motion Analysis), Altair EDEM (Simulation)

2. Medical Device Development — Controlled Flow Mechanical System

Industry: Medical Technology

Key Contributions:

- Designed mechanical components for controlled flow and operational safety.
- Developed assemblies involving valves, filters, actuators, and flow-control components.
- Applied mechanical design principles related to pressure management, thermal considerations, and fluid behavior.
- Implemented DFM and DFA methodologies for manufacturing readiness.
- Performed simulation-based design validation.
- **Challenges Managed:** Maintaining reliable mechanical operation under strict performance requirements. Improved design reliability through engineering analysis and simulation.
- **Tools Used:** SolidWorks (CAD Modeling & Motion Analysis), ANSYS (Simulation & Validation)

3. Design and Fabrication of Mechanism for Automated Installation of 18 Liter water bottle of Water Dispenser

Industry: Academic Research (Final Year Project)

Key Contributions:

- Designed the complete 3D CAD model for the automated mechanism.
- Developed a comprehensive solution to automate the manual lifting process, significantly reducing human effort.
- Engineered a customized lifting-to-adjustment sequence for seamless water bottle installation and atomization.
- Optimized the design by utilizing a minimum number of mechanisms to achieve cost-effective and reliable results.
- **Challenges Managed:** Designing a reliable lifting mechanism capable of handling 18-liter loads safely and efficiently. Balancing cost-effectiveness with robust mechanical performance through design optimization.
- **Tools Used:** SolidWorks (3D Modeling, Simulation, Animation, Motion Analysis)

LANGUAGES

- **English** (Professional) IELTS Band 6.0
- **Seraiki** (Native)
- **Urdu & Hindi** (Fluent)

CERTIFICATIONS

- Pakistan Engineering Council (PEC), Pakistan
- Certified SOLIDWORKS CAD Design Associate
- Lean Six Sigma White Belt and White Belt (LSSWB and LSSYB)
- Lean Fundamentals and Production
- Data Analytics in Lean Six Sigma
- CPR, AED and First Aid

REFERENCES

Professor Dr. Naveed Ahmad

- Assistant Professor, UET, Peshawar, Khyber Pakhtunkhwa, Pakistan
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